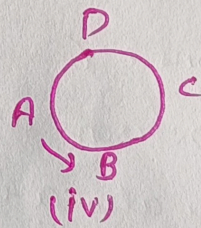
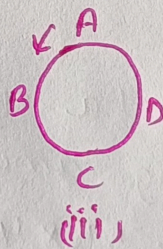
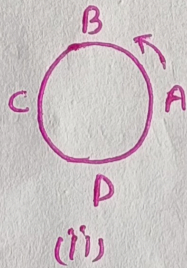
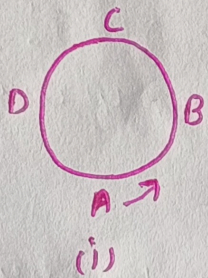


CIRCULAR PERMUTATION

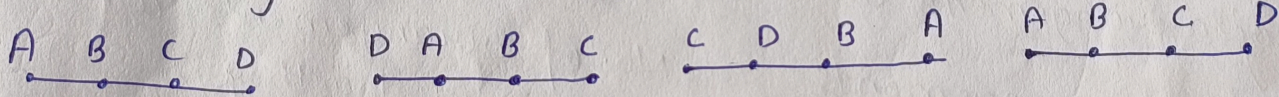
The permutations till now were linear permutations as the objects are being arranged in a row.

If we arrange them in a circle, then the no. of permutations change. A circular permutation is an arrangement of objects around a circle.

Consider 4 persons A, B, C and D who are to be arranged around a circle. The following arrangements are one and the same as relative position of the four positions A, B, C, D is changed.



But in case of linear arrangements.



As the no. of circular permutations depend on the relative positions of the objects, the no. of permutation of n objects in a circle can be found by keeping one element fixed and arranging remaining $(n-1)$ elements which can be done in $(n-1)!$ ways.

The number of circular arrangements of n different things
 $= (n-1)!$

Q In how many ways 5 boys and 4 girls can be seated at a round table?

- ① there is no restriction
- ② all four girls sit together
- ③ all four girls do not sit together.
- ④ no two girls sit together.

① Total no. of persons = $5+4=9$
Persons can be seated on a round table in $(9-1)!$ ways = $8!$ ways.

② Taking 4 girls as 1.

No. of persons = 6

Sitting of 6 persons around a round table is $(6-1)! = 5!$ ways

But 4 girls can also arrange among themselves in $4!$ ways.

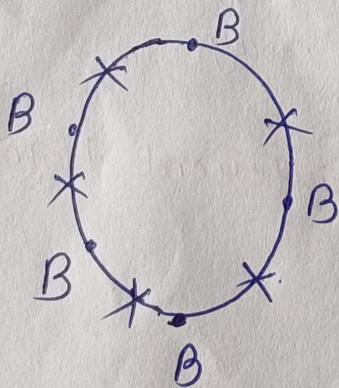
So, required no. of ways = $5! \times 4!$

③ No. of arrangements when all 4 girls do not sit together

= Total no. of arrangements without restriction — No. of arrangements when all 4 girls sit together

$$= 8! - 5! \times 4!$$

iv)



There is no restriction on Boys.

So, first we arrange 5 boys in $(5-1)! = 4!$ ways

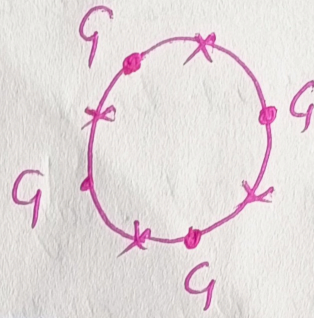
Now, there are 5 places for girls as indicated by X.

$\therefore$ 4 girls can be seated in ${}^5P_4 = 5!$ ways

So, required no. of ways = $5! \times 4!$

Q In how many ways can a party of 4 boys and 7 girls can be seated at a circular table so that no 2 boys are adjacent.

4 girls can be seated around a round table in $(4-1)! = 3!$ ways.



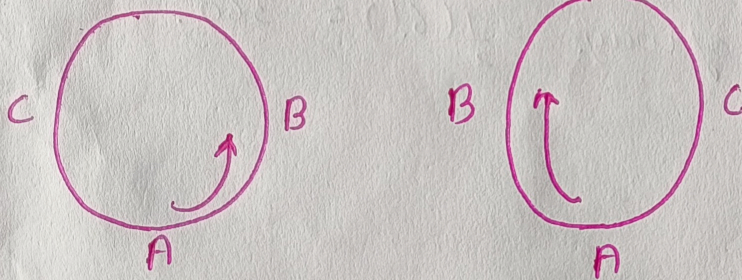
For Boys, remain 4 places, so they can be seated in $4!$ ways.

Thus, there are $3! \times 4!$ i.e. 144 ways.

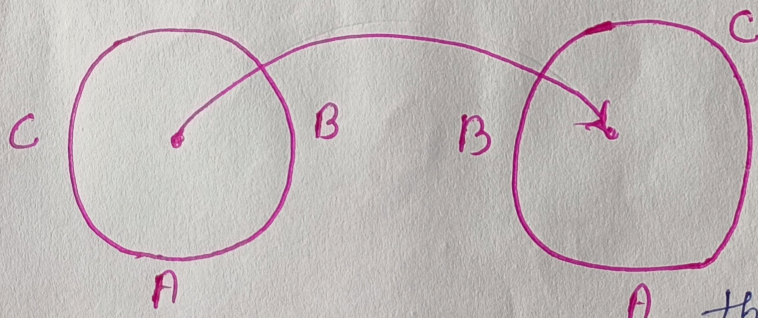
Clockwise and Counter-Clockwise Permutations

There are two types of circular permutations.

(i) Those permutations in which clockwise and anti-clockwise arrangements are distinguishable. Thus, while seating 3 persons A, B, C around a table, the following permutations are different.



(ii) When clockwise and anti-clockwise permutations are not different.



Ex: arrangement of beads in a necklace, arrangement of flowers in a garland etc., no. of circular arrangements of n diff.

things is $\frac{1}{2}(n-1)!$

Q In how many ways 8 different beads can be arranged to form a necklace?

Eight different beads can be arranged in circular form in $(8-1)! = 7!$ ways.

Since, there is no distinction b/w clockwise and anti-clockwise arrangements, the required no. of arrangements =

$$\frac{1}{2} (8-1)! = \frac{1}{2} (7!)$$

Q In how many ways can seven persons sit around a table so that all shall not have the same neighbours in any two arrangements?

Seven persons can sit around a table in $(7-1)! = 6!$ ways.

But in clockwise and anti-clockwise arrangements, each person will have same neighbours.

So, required no. of ways = $\frac{1}{2} (6!) = 360$.

Combinations

No. of combinations of 'n' different things taken 'r' at a time is given by

$${}^nC_r = \frac{n!}{r!(n-r)!}$$

Eg How many poker hands of 5 cards can be dealt from a standard deck of 52 cards?

Solⁿ $n = 52$ $r = 5$

$${}^{52}C_5 = \frac{52!}{5!(52-5)!}$$

$$= \frac{52!}{5!(47)!}$$

Eg Prove that ${}^nC_r = {}^nC_{n-r}$

Solⁿ ${}^nC_r = \frac{n!}{r!(n-r)!}$

$${}^nC_{n-r} = \frac{n!}{(n-r)![n-(n-r)]!}$$

$$= \frac{n!}{(n-r)!(r)!}$$

Hence ${}^nC_r = {}^nC_{n-r}$

Combinations with repetitions

No. of ways to choose 'r' objects from a collection of 'n' objects with repetition allowed is

$$C(n+r-1, r)$$

Q A man wants to buy some pet birds. The pet store sells parrots, sparrows & peacocks. How many different selections are possible if the man wants to take home 7 birds.

Solⁿ $n = 7$ $r = 3$ } Some birds are repeated

$$C(n+r-1, r) = C(3+7-1, 7)$$

$$= C(9, 7)$$

$$= \frac{9!}{7! \times 2!} = \boxed{36}$$

Important Combinations formulas

1) No. of combinations of 'n' different things taken 'r' at a time when 'p' particular things are always included.

$${}^{n-p}C_{r-p}$$

2) No. of combinations of 'n' different things taken 'r' at a time when 'p' particular things are always to be excluded :- ${}^{n-p}C_r$

Q In how many ways can a cricket 11 be chosen out of 15 players ? If

- i) a particular player is always chosen
- ii) a particular player is never chosen

Solⁿ $n = 15$ $r = 11$

(i) Always included

$$p = 1 \quad n-p \text{ C } r-p \text{ ie } {}^{15-1}C_{11-1} \Rightarrow {}^{14}C_{10}$$

$$\frac{14!}{10! 4!} = \textcircled{1365}$$

$$\text{ii) } n-p \text{ C } r \Rightarrow {}^{15-1}C_{11} \Rightarrow \frac{14!}{11! 3!} = \textcircled{364}$$